

Revisiting Cross-Listing: Examining Stock Returns and Market Integration

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Abstract

This study revisits the impact of cross-listing on major U.S. exchanges on financial integration, stock returns, and risk profiles of international firms and investigates the return reactions of peers and the strategic adaptations they undertake in response to cross-listing. Using **institutional and competitive dynamics theories**, the research explores how enhanced legitimacy, reduced information asymmetry, and peer signaling affect firms' financial outcomes post-cross-listing. The sample includes 1,142 cross-listed firms from 44 countries from 1990 to 2023. Employing event study methodology, the findings reveal that cross-listing lowers firms' cost of capital, resulting in reduced expected returns and negative abnormal returns in the short and medium term after the announcement. Firms from non-segmented markets benefit more from cross-listing, experiencing lower information asymmetry and greater alignment with international standards. Peer effects show that the cross-listing of firms influences the abnormal returns of their peers, who adjust their strategies to compete more effectively within the new, integrated market environment. This adaptation leads to improved financial performance for peers that successfully mirror the cross-listed firm's strategies over the long term.

Keywords: Cross-listing, Institutional theory, Competitive dynamics theory, Abnormal returns, Strategic adaptation

A) Introduction

The phenomenon of cross-listing, where companies list their shares on foreign stock exchanges while maintaining a primary listing in their home country, has been a subject of significant academic and practical interest for decades (Alexander, Eun, & Janakiraman, 1988; Karolyi, 1998). This practice has come under renewed scrutiny due to its potential implications for global financial integration, stock returns, and corporate financial strategy. Cross-listing enables firms to access larger capital markets, enhance liquidity, and attract a broader investor base (Foerster & Karolyi, 1998; Doidge, Karolyi, & Stulz, 2004). Recent studies suggest that cross-listing trends have evolved significantly, with an increasing number of firms leveraging this strategy to compete in an interconnected global economy (Gozzi, Levine, & Schmukler, 2008). However, whether cross-listing meaningfully contributes to the integration of capital markets remains a fundamental question for international finance scholars and policymakers.

Existing literature has explored various dimensions of cross-listing, including its impact on firm valuation, investor base expansion, and liquidity. Alexander, Eun, and Janakiraman (1988) pioneered the empirical investigation of international dual listings, identifying stock price changes associated with this practice as indicative of market segmentation. Subsequent studies by Foerster

and Karolyi (1998) and Miller (1999) provided further evidence on the role of liquidity and investor recognition in influencing returns, while examining institutional differences across markets. More recent research has expanded on these themes, incorporating data from the post-2000 era to reflect changes in global financial dynamics. For instance, Doidge, Karolyi, and Stulz (2004) explored how legal environments and investor protection affect cross-listing outcomes, while Gozzi, Levine, and Schmukler (2008) analyzed the impact of financial globalization on market integration. Studies by King and Segal (2009) and Sarkissian and Schill (2012) have also investigated the evolving barriers between segmented and integrated markets, highlighting the roles of technological advancements and regulatory changes. These findings suggest that the effects of cross-listing extend beyond simple access to foreign capital markets, shedding light on both enduring and emerging barriers to market integration.

Despite previous research into cross-listing, several critical questions remain underexplored or warrant revisiting due to the evolving dynamics of financial markets. One key area of interest is understanding the behavior of stock returns before and after the announcement of a cross-listing, as well as the behavior of returns for firms in segmented versus non-segmented markets around these announcements. These areas are crucial because they provide valuable insights into how cross-listing influences a firm's financial performance and market perception. Examining the behavior of returns can reveal the extent to which cross-listing impacts stock liquidity, investor recognition, and risk exposure, all of which are fundamental for assessing a firm's competitiveness and integration into global markets. Earlier studies, such as those by Foerster and Karolyi (1999) and Miller (1999), explored these questions and found that cross-listing was associated with significant changes in stock returns, particularly demonstrating the presence of market segmentation. They found that cross-listing led to positive abnormal returns before the announcement and negative abnormal returns afterward, indicating that market participants adjusted to the new level of integration. This pattern suggests that while cross-listing initially enhanced investor recognition and market access, it also led to market revaluation after the cross-listing event. This shift implies that cross-listing alone may not fully integrate firms into global markets; rather, the market's reaction highlights the presence of ongoing barriers to integration. The negative post-announcement returns reflect the market's adjustment to the increased exposure and risk that come with a new level of integration, challenging the notion that cross-listing alone provides a straightforward path to greater financial integration. However, these studies were limited in scope, relying on data primarily from the 1990s and focusing on a narrow set of firms and countries. As financial markets have become more integrated since the early 2000s, revisiting these findings with more comprehensive and updated datasets could provide new insights into how segmented and non-segmented markets react differently to cross-listing events.

An additional critical area that needs to be revisited is the behavior of the market risk profile of cross-listed firms in relation to their home country's market both before and after the cross-listing, as well as in relation to global markets. This is important because understanding how cross-listing affects a firm's risk exposure in these contexts can provide valuable insights into its integration with international financial systems and its overall market positioning. Analyzing these dynamics

will help to reveal the extent to which cross-listing influences a firm's risk profile, either by enhancing its stability and liquidity due to broader investor recognition and access to global capital, or by exposing it to new types of risk associated with international operations and regulatory environments. Recent research by Bae, Stulz, and Tan (2008) and Doidge et al. (2017) has examined post-2000 cross-listing trends, illustrating changes in firm risk profiles and the integration of global financial markets. For instance, Doidge et al. (2017) found that the risk reduction benefits of cross-listing have diminished over time, possibly due to enhanced investor recognition and liquidity across markets. Moreover, the role of cross-listing in altering a firm's risk exposure, particularly as measured by market beta, remains an area deserving further exploration. Earlier studies suggested that cross-listing might lower risk due to enhanced visibility, liquidity, and investor diversification, as seen in the work of Acharya, Johnson, and Tiran (2009). However, these conclusions were drawn under market conditions that no longer fully represent today's global financial landscape. Modern cross-listing dynamics, influenced by technological advancements and regulatory changes, could significantly alter a firm's exposure to market risks. For example, the increased access to real-time financial data and rapid dissemination of information has heightened market efficiency, making the role of market beta even more critical in evaluating a firm's risk exposure. Understanding these evolving dynamics is crucial for both academics and practitioners aiming to gain a deeper understanding of how cross-listing affects a firm's risk profile in today's interconnected financial environment.

In addition to revisiting these questions, a key unexplored area demands greater attention: how local peers react to cross-listing announcements, both in the short and long term. This question is critical for understanding the broader implications of cross-listing on market integration. For instance, a cross-listing may signal increased competition or serve as a benchmark for local firms, influencing their valuation, strategic decisions, or risk profiles. Understanding the peer reaction is vital for assessing how cross-listing influences market efficiency and financial integration. When local peers react positively to a cross-listing announcement, it can enhance the overall market sentiment, reduce information asymmetry, and increase liquidity across the sector. This ripple effect not only benefits the individual cross-listed firm but also fosters a more integrated financial market environment, where other firms are incentivized to adopt similar strategies. By aligning with global financial standards and practices, local firms may enhance their market integration, paving the way for more seamless capital flows and a higher degree of financial interconnectivity. Therefore, the peer reaction is an important indicator of the willingness and capacity of local markets to engage more deeply at a global level, thus amplifying the benefits of cross-listing for both individual firms and the broader market ecosystem. The local peer effect is particularly important in highly interconnected markets, where the actions of one firm can ripple across an industry. Research has shown that spillover effects from significant market events, such as acquisitions or initial public offerings (IPOs), can influence the behavior and performance of other firms within the same sector (e.g., Spiess and Affleck-Graves, 1995; Loderer and Martin, 1997). By examining the reactions of local peers, researchers can gain deeper insights into how cross-listing shapes competitive dynamics, information asymmetry, and capital flows at a regional level.

This analysis is vital for policymakers and market participants aiming to foster efficient and integrated financial markets. For instance, studies on peer effects in IPO markets demonstrate how the introduction of a cross-listed firm can lead to spillover benefits, such as enhanced market liquidity and reduced information asymmetry for other firms in the industry (Chemmanur et al., 2003). Understanding these dynamics is essential not only for the firms themselves but also for broader financial stability, as it can help identify the channels through which cross-listing influences market behavior and regulatory policies.

In summary in this paper, I aim to revisit and address the three main questions explored in previous studies related to cross-listing. First, I will examine changes in abnormal stock returns before and after the cross-listing announcement to gain a deeper understanding of how market participants adjust to the new level of integration. Second, I will analyze the behavior of firms from segmented versus non-segmented markets around cross-listing events to shed light on the challenges and implications of cross-listing for market integration. Third, I will investigate changes in a firm's market beta relative to its home and international markets, which can provide insights into the evolving risk exposure associated with cross-listing. Additionally, I will focus on an often-overlooked aspect of cross-listing—the reaction of local market peers to the cross-listing announcement, both in the short and long term.

To address the research questions posed in this study, I will employ an event study methodology. This approach is suitable for capturing the market's reaction to cross-listing announcements. The event study methodology allows for the examination of abnormal returns surrounding the cross-listing event, providing insights into how the market adjusts its expectations in response to new information. Specifically, I will utilize short-term event windows (such as 11-day, 5-day, and 3-day) to capture immediate market reactions and long-term windows (1-year, 2-year, and 3-year) to assess sustained effects on stock performance and risk profiles. By analyzing these windows, I aim to offer a comprehensive understanding of how cross-listing influences firm returns, market integration, and risk dynamics over different time horizons.

In summary, this paper revisits the core questions of previous studies on cross-listing, examining how it influences stock returns, market risk profiles, and peer reactions in both the short and long term. The methodology chosen—an event study using multiple time windows—will enable a thorough exploration of these issues, shedding light on the broader implications of cross-listing for financial integration and market dynamics. The findings from this study will not only contribute to the existing literature but also provide valuable insights for policymakers and firms considering cross-listing as a strategy for enhancing their global competitiveness. The following sections of this paper will present a detailed analysis of these issues, supported by empirical evidence and a discussion of their implications for international finance.

B) Research Context

I chose companies that have engaged in cross-listing on major U.S. exchanges, including NYSE, AMEX, and NASDAQ, between 1990 and 2023 as the research subjects and the U.S. stock exchanges as the foreign capital market in my main data analysis primarily for two reasons: (1)

the significant increase in the number of cross-listed firms post-1990 on major U.S. exchanges, including NYSE, AMEX, and NASDAQ, reflecting the growing trend of global financial integration. This trend is driven by multinational companies seeking to enhance their market profile, access larger pools of capital, and achieve greater investor recognition (Foerster & Karolyi, 1998; Doidge, Karolyi, & Stulz, 2004). (2) The U.S. exchanges offer mature regulatory frameworks, deep liquidity, and a broad investor base, making them increasingly attractive to firms worldwide. This choice reflects a strategic shift by companies to leverage the benefits of financial globalization and enhance their competitiveness on a global scale (Gozzi, Levine, & Schmukler, 2008). Figure 1 illustrates the dynamic trend of cross-listed firms on major U.S. exchanges, including NYSE, AMEX, and NASDAQ, from 1990 to 2023 on a monthly scale. Initially, there were 76 companies cross-listed in January 1990. The number increased steadily to a peak of 358

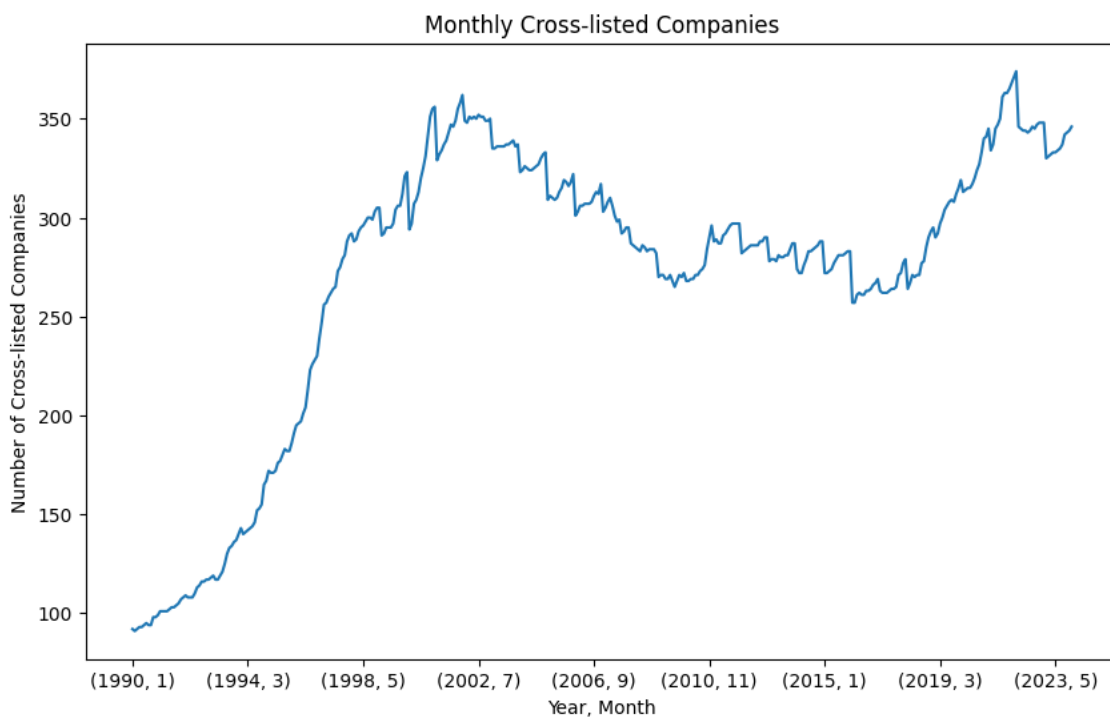


Figure 1. trend of cross-listed firms on major U.S. exchanges, including NYSE, AMEX, and NASDAQ.

by July 2002. However, by May 2015, there was a decline to 264 cross-listed firms. Despite this decrease, by 2023, the trend rebounded with 350 companies cross-listed, indicating renewed interest and the evolving strategic choices of firms seeking to enhance their market profiles.

Despite the higher costs and challenges of listing and renewing listings in foreign markets (Xu, 2014), many companies prefer the U.S. and other developed countries' stock markets over their domestic ones for several reasons. Beyond the common disadvantages faced by most developing countries—such as immature stock markets that provide less capital and a narrower base of shareholders, which limit companies' long-term growth (Ding et al., 2010; Foerster & Karolyi, 1999; Pagano et al., 2002), and underdeveloped legal and financial systems that create higher risks for issuers and investors (Allen, Qian, & Qian, 2005; Cumming, Fleming, & Schwienbacher, 2006, 2009; La Porta, Lopez-de-Silanes, Shleifer, & Vishny, 1998)—there are additional institutional

features of non-U.S. markets that make many non-U.S. entrepreneurs choose to list on well-developed markets.

First, non-U.S. markets often have stringent criteria for companies to list, such as minimum profitability or asset thresholds, which can be particularly challenging for new issues, especially in high-growth sectors like IT and Internet (Xu, 2014). These requirements may be less flexible compared to the U.S. market. Second, U.S. markets offer a more streamlined process for IPOs, which is more conducive to rapid fundraising (Ding et al., 2010). Third, industries in non-U.S. markets, especially those related to technology and education, face restrictions on foreign ownership. By listing in the U.S., these firms can bypass such restrictions and attract necessary investments (Pearson, 2012).

After companies decide to seek cross-listing abroad, they must select the most suitable listing market (Moore et al., 2012; Sarkissian & Schill, 2004). A significant number of companies opt for the United States due to several compelling reasons.

Besides the general reasons such as the home and host country legal regimes and cultural influences (e.g., Bell, Moore, et al., 2012; Cumming et al., 2006, 2009), several specific factors influence companies' decisions to cross-list on major U.S. exchanges.

First, the unique listing requirements of U.S. exchanges are particularly conducive to pre-profit high-tech new ventures (Xu, 2014). The NYSE and NASDAQ offer flexible options for listing based on assets, shareholders' equity, and market capitalization alone, which allows these companies to access capital more easily. Second, the U.S. stock market is the largest and arguably the most efficient in the world (Hursti & Maula, 2007). This efficiency enables companies to obtain more capital with relatively low levels of underpricing and risks during the cross-listing process, which is crucial for maintaining competitive advantage and managing market risk profiles (Aggarwal et al., 2011; Bell, Filatotchev, et al., 2012; Ding et al., 2010; Moore et al., 2012). Third, cross-listing on U.S. exchanges enhance companies' prestige, thereby increasing their competitive advantages in both domestic and foreign markets over the long run. This is particularly significant for firms seeking to integrate into global financial markets and build investor recognition (Aggarwal et al., 2011; Ding et al., 2010; Moore et al., 2012).

So far, we have explored the reasons companies prefer to cross-list on U.S. exchanges. Now, we turn our attention to the data to examine the trends in cross-listing at the country level. Figure 2 illustrates the trends in the number of cross-listed firms from the top 20 countries with the highest cross-listing activity. This analysis highlights the variation in cross-listing patterns across different countries, providing deeper insights into how firms from diverse economic and regulatory environments have engaged with U.S. capital markets over time.

Figure 2 presents a heat map depicting the number of firms from various countries that have experienced cross-listing on U.S. exchanges between 1990 and 2023. For visual clarity, the numbers have been transformed using a logarithmic scale. The heat map reveals significant trends at the country level, with firms from China exhibiting the most substantial increase in cross-listings—from just 3 firms in 1990 to 146 in 2023. Similarly, countries such as Argentina, Brazil, Israel, and South Korea also demonstrate notable growth in the number of cross-listed firms over

the study period, reflecting the increasing globalization of capital markets and the strategic appeal of U.S. exchanges for firms worldwide.

This analysis underscores the importance of revisiting the central questions explored in previous studies on cross-listing. The dynamic trends observed, particularly the substantial increase in cross-listed firms from countries such as China, Brazil, and South Korea, highlight the evolving nature of cross-listing and its implications for market integration, stock returns, and risk dynamics. These trends not only reaffirm the significance of understanding the abnormal stock returns and market integration effects associated with cross-listing but also introduce new dimensions to consider.

The growing number of firms cross-listing from emerging markets, coupled with the strategic motivations driving these decisions, brings renewed attention to the peer effects of cross-listing announcements. Understanding how local market peers respond—both in the short and long term—offers valuable insights into the broader market impacts of cross-listing.

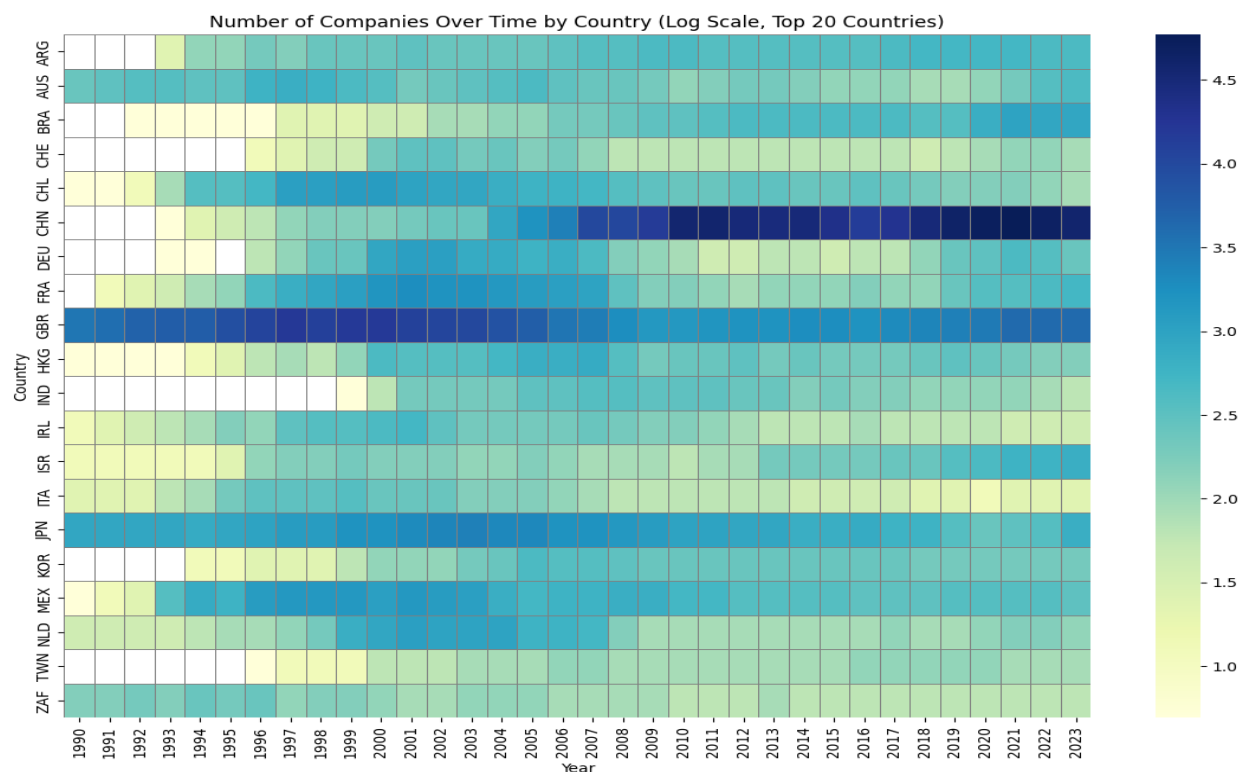


Figure 2. Trends in cross-listing at the country level

By revisiting these foundational questions and incorporating the changing dynamics at the country level, this research aims to contribute to the broader understanding of cross-listing and its implications in a rapidly globalizing financial landscape. The increasing participation from diverse countries not only validates the need for such studies but also emphasizes the relevance of examining these phenomena with fresh perspectives and updated methodologies.

The research context underscores the critical role of U.S. stock exchanges in facilitating global financial integration and providing firms with access to unparalleled resources. Between 1990 and

2023, these exchanges witnessed a dynamic shift in cross-listing activity, reflecting changes in the global financial landscape and the strategic priorities of firms worldwide. This timeframe captures the acceleration of globalization, the rise of emerging markets, and pivotal regulatory developments, such as the introduction of the Sarbanes-Oxley Act, which temporarily affected cross-listing trends (Litvak, 2007; Doidge et al., 2009).

While firms from emerging markets like China and Brazil have dominated the recent surge in cross-listings, companies from developed markets, including the United Kingdom and Canada, have also maintained strong participation (Pagano et al., 2002; Karolyi, 2006). These trends reflect varying motivations, from accessing capital in more liquid markets to leveraging the U.S.'s advanced regulatory framework for investor protection and transparency (Coffee, 2002; Foerster & Karolyi, 1998).

Cross-listing on U.S. exchanges not only provides firms with direct benefits, such as enhanced liquidity and investor base expansion, but also has broader implications for market integration. Firms from segmented markets often exhibit more pronounced changes in stock returns and market risk profiles post-listing, as they transition to a more integrated financial environment (Stulz, 1999; Karolyi & Stulz, 2003). The dynamic behavior of local market peers in response to cross-listing events further underscores the importance of examining peer effects, which remain a relatively underexplored dimension in cross-listing research (King & Segal, 2009; Lel & Miller, 2008).

This study builds on these observations to revisit foundational questions about abnormal stock returns, market integration, and risk dynamics. By incorporating trends at the country level and analyzing the evolving motivations for cross-listing, it aims to provide fresh insights into how globalization reshapes the competitive and financial strategies of firms. The substantial increase in cross-listing activity, particularly from emerging markets, further emphasizes the need to understand how local and global markets respond to these strategic decisions. These trends align with earlier studies that highlight the increasing importance of cross-border financial activities in shaping firm valuation, market behavior, and investor recognition (Doidge, Karolyi, & Stulz, 2004; Gozzi, Levine, & Schmukler, 2008).

C) Theory Development and Hypotheses

This research draws on **institutional theory** to address the first three research questions and **competitive dynamics theory** to examine the peer effects of cross-listing.

Institutional theory has long contended that organizations are embedded in broader institutional environments where norms, values, and beliefs significantly shape their strategies and behaviors (DiMaggio & Powell, 1983; Meyer & Rowan, 1977). Institutional theory is particularly relevant in explaining strategic decisions that extend beyond purely rational or economic calculations (DiMaggio, 1988). Cross-listing, as a strategic move, often reflects firms' attempts to achieve **legitimacy**—a central concept in institutional theory. Legitimacy is socially constructed and conferred when organizations align with the norms and expectations of key stakeholders, such as investors and regulators (Dowling & Pfeffer, 1975; Oliver, 1991).

In the context of cross-listing on U.S. exchanges, legitimacy can be viewed as a mechanism through which firms signal their adherence to superior governance standards and financial transparency to global stakeholders (Doidge, Karolyi, & Stulz, 2004; Siegel, 2005). This is particularly critical given the liability of foreignness and information asymmetries that non-U.S. firms face when entering U.S. markets (Zaheer & Mosakowski, 1997). Firms must overcome barriers such as unfamiliarity with U.S. listing procedures, cultural differences, and stringent regulatory frameworks. Cross-listing, however, allows firms to address these challenges by aligning with the robust institutional structures of U.S. markets, enhancing their perceived efficiency, competence, and trustworthiness (Deeds, Mang, & Frandsen, 2004; Grinblatt & Keloharju, 2000).

For the analysis of peer effects, **competitive dynamics theory** offers a framework, emphasizing the strategic interactions among firms operating within competitive environments (Chen & Miller, 1994; Ferrier, Smith, & Grimm, 1999). Cross-listing by a firm represents a strategic move that can disrupt the competitive equilibrium within its home market. By gaining increased access to capital, enhancing global visibility, and signaling superior competitiveness, cross-listed firms may exert pressure on their local peers to respond strategically. Such responses might include changes in financial policies, operational strategies, or even attempts to pursue their own cross-listing opportunities. Competitive dynamics theory thus provides valuable insights into the ripple effects triggered by cross-listing, capturing both the direct and indirect impacts on market rivals.

C1.Behavior of stock returns around the cross-listing announcement

Cross-listing on major U.S. exchanges is often regarded as a strategic commitment by firms to enhance corporate governance and transparency, attributes highly valued by institutional investors (Doidge, Karolyi, & Stulz, 2004; Foerster & Karolyi, 1998). According to **institutional theory**, legitimacy plays a critical role in how external stakeholders, particularly institutional investors, evaluate firms. By cross-listing on U.S. exchanges, firms signal their adherence to the higher standards of governance, compliance, and financial reporting that are characteristic of these markets (DiMaggio & Powell, 1983; Meyer & Rowan, 1977).

This commitment is interpreted by the market as a sign of improved operational and financial quality, leading to enhanced investor perceptions and expectations of future performance. Consequently, investors revise their valuation models to reflect this newfound legitimacy, resulting in market reactions around the time of the cross-listing announcement. The increased legitimacy and anticipated reduction in information asymmetry are expected to narrow the gap between expected and realized returns. In some cases, this alignment of expectations may lead to **negative abnormal returns** if the market has already anticipated the benefits of cross-listing, or if the firm's value was previously overestimated. **Abnormal returns** are calculated as the difference between the actual return and the expected return based on historical data and market conditions (Aggarwal et al., 2011; Karolyi, 2006).

Empirical studies provide mixed evidence on the behavior of abnormal returns around cross-listing announcements. Some studies document significant positive abnormal returns, reflecting the

market's perception of increased transparency and legitimacy (Foerster & Karolyi, 1998; Siegel, 2005). However, other research suggests that if investors had already priced in the expected benefits of cross-listing, abnormal returns might decrease or even become negative, indicating that the announcement reduced uncertainty rather than provided new, value-enhancing information (Doidge, Karolyi, & Stulz, 2004).

Before cross-listing, firms typically face higher levels of information asymmetry and lower legitimacy, leading to greater discrepancies between expected returns and realized returns. After cross-listing, the enhanced legitimacy and reduced information asymmetry help align market expectations with actual returns, which could result in a reduction—or even reversal—of abnormal returns. As a result:

***Hypothesis 1a:** Before the announcement, firms face high information asymmetry and low legitimacy, leading to higher expected returns, potentially resulting in positive abnormal returns.*

***Hypothesis 1b:** After the announcement, the gained legitimacy and reduced information asymmetry lead to a closer alignment between expectations and actual returns, which may result in lower and negative abnormal returns after the announcement.*

C2.Segmented markets vs non-segmented markets

Segmented markets are characterized by institutional barriers such as regulatory constraints, informational asymmetry, and restricted access to global resources (Hoskisson et al., 2000; Li, 2003). These barriers give rise to a “liability of foreignness” (Zaheer, 1995), where firms from segmented markets struggle to gain legitimacy and acceptance in international financial arenas. Firms in these markets often face higher costs and administrative burdens, including compliance with complex and inconsistent regulatory frameworks, differing disclosure requirements, and managing investor skepticism about their governance practices and financial transparency (Certo, 2003).

To classify countries as segmented or non-segmented, this study employs the Chinn-Ito Financial Openness Index (**KAOPEN**), a widely recognized measure of a country's degree of capital account openness (Chinn & Ito, 2006). The KAOPEN index is derived from International Monetary Fund (IMF) data and is constructed based on the presence of multiple types of capital controls, such as restrictions on capital inflows and outflows.

When firms from segmented markets decide to cross-list on major U.S. exchanges, they signal a strategic commitment to align with superior governance and transparency standards. This decision is interpreted by the market as an effort to overcome the institutional barriers of their home markets. Institutional investors, particularly sensitive to risks associated with information asymmetry and governance quality, view such actions as mitigating these risks. Consequently, this perception influences the market's evaluation of the firm's legitimacy and expectations about its future performance (Doidge, Karolyi, & Stulz, 2004; Foerster & Karolyi, 1998).

C2-1 Before Cross-Listing: Segmented Markets and the Expectations Gap

Prior to cross-listing, firms from segmented markets face higher levels of information asymmetry and lower legitimacy, creating a substantial gap between expected and realized returns. This gap is primarily due to investor uncertainty about the firm's governance practices, operational quality, and ability to adhere to international standards. In contrast, firms from non-segmented markets operate in environments with fewer institutional barriers and more established reputations, resulting in a smaller gap between expectations and realized returns.

***Hypothesis 2a:** Before the cross-listing announcement, firms from segmented markets experience a larger gap between expected and realized returns (positive abnormal returns) compared to firms from non-segmented markets, due to greater information asymmetry and lower legitimacy.*

C2-2 After Cross-Listing: Alignment of Expectations and Realized Returns

Cross-listing enables firms from segmented markets to reduce information asymmetry and enhance legitimacy by adhering to the governance and disclosure standards of U.S. exchanges. This leads to a closer alignment between investor expectations and actual performance. Consequently, the **abnormal returns** of these firms decrease after cross-listing, reflecting the reduced gap between perceived risk and actual risk.

In contrast, for firms from non-segmented markets, the cross-listing process does not produce the same level of incremental legitimacy or reduction in information asymmetry, as these firms are already operating in more transparent and integrated financial systems. Therefore, their abnormal returns are expected to remain relatively unchanged after cross-listing.

***Hypothesis 2b:** After the cross-listing announcement, firms from segmented markets experience a decrease in abnormal returns (negative abnormal returns) due to the alignment of expectations with actual performance, while firms from non-segmented markets exhibit no significant change in abnormal returns.*

C3. Risk Exposure (Market Beta)

The cross-listing of firms on major U.S. exchanges significantly impacts their market risk profile, particularly their beta. Beta measures a firm's sensitivity to market movements and provides insights into its risk exposure relative to local and global markets. Institutional theory offers a useful framework to understand these changes, as cross-listing can influence a firm's risk profile through enhanced legitimacy, broader investor recognition, and integration into global financial systems (DiMaggio & Powell, 1983; Meyer & Rowan, 1977).

Before cross-listing, firms' beta is predominantly tied to their local market due to high information asymmetry and limited investor base. Firms in segmented markets often face significant institutional barriers, such as weak enforcement of corporate governance and inadequate transparency. These conditions lead to greater dependency on local market factors and higher beta relative to the local market. In contrast, the firm's beta relative to global (U.S.) markets is typically low due to minimal integration with international financial systems (Karolyi, 2006). This scenario reflects the firm's limited exposure to global economic factors and its reliance on domestic economic conditions for valuation (Doidge, Karolyi, & Stulz, 2004).

Cross-listing alters this dynamic by reducing information asymmetry and improving transparency through adherence to U.S. regulatory and governance standards. These changes enhance the firm's legitimacy and credibility among global investors, resulting in increased access to international capital and greater market integration (Bae, Stulz, & Tan, 2008; Acharya, Johnson, & Tiran, 2009). The shift in investor base from predominantly local to more diversified global participants diminishes the firm's dependency on local markets, leading to a reduction in its beta relative to the local market. At the same time, the firm becomes more exposed to global market movements, increasing its beta relative to the U.S. market.

Post-cross-listing, the firm's overall beta often moderates due to enhanced visibility and liquidity, as well as the broader diversification of its investor base. These factors stabilize the firm's risk profile, though the degree of beta adjustment may vary depending on the firm's industry, market conditions, and timing of cross-listing. Recent studies, such as those by Doidge et al. (2017), indicate that while cross-listing historically reduced firms' risk exposure, its effects have evolved in today's interconnected financial environment. Technological advancements and increased market efficiency, such as real-time financial data access, have heightened the sensitivity of beta to global risk factors.

Additionally, the global nature of U.S. exchanges exposes cross-listed firms to regulatory and macroeconomic risks inherent in the host market. This exposure can elevate the beta related to U.S. markets, reflecting the firm's deeper integration into international financial systems (Acharya, Johnson, & Tiran, 2009). Understanding these dynamics is crucial for academics and practitioners aiming to evaluate how cross-listing affects a firm's stability, liquidity, and investor perception in modern financial markets.

Hypothesis 3a: *The beta related to the local market decreases after cross-listing as the firm's operations and risk profile become less dependent on local market conditions.*

Hypothesis 3b: *The beta related to the U.S. market increases after cross-listing as the firm's integration with global markets and exposure to global risk factors grows.*

C4.What are the peer effects?

The cross-listing of a firm on a major U.S. exchange can create significant ripple effects among its peers in the home market. These effects can be analyzed using competitive dynamics theory, which examines how firms within the same competitive environment respond to the strategic actions of others (Chen & Miller, 1994). Cross-listing represents a strategic move that not only enhances the legitimacy and financial standing of the cross-listed firm but also signals heightened competitive pressure to its peers. These peers, in turn, must evaluate and adapt their strategies to maintain their market position.

Peers may adopt various strategies to mitigate the competitive advantage gained by the cross-listed firm. These strategies reflect efforts to maintain competitiveness in a changing landscape influenced by globalization and heightened investor expectations. One significant approach is improving corporate governance and transparency. Peers may enhance their financial disclosures by increasing both the quality and frequency of reporting, including adopting practices such as

quarterly reporting, providing detailed earnings guidance, and integrating Environmental, Social, and Governance (ESG) metrics into their reports (La Porta et al., 1998; Aguilera & Jackson, 2003). Additionally, they might restructure their boards to include a higher proportion of independent directors, ensuring stronger oversight and reducing potential conflicts of interest. To further demonstrate a commitment to corporate integrity, peers may voluntarily align their operations with globally recognized frameworks such as the International Financial Reporting Standards (IFRS) or adhere to corporate governance codes like those issued by the OECD (Claessens & Yurtoglu, 2013).

Another key strategy involves seeking internationalization. Inspired by the success of the cross-listed firm, peers may explore opportunities to pursue their own cross-listing on major international exchanges, such as the NYSE, NASDAQ, or the London Stock Exchange (Doidge, Karolyi, & Stulz, 2004). Alternatively, they may engage in strategic alliances by forming partnerships or joint ventures with international companies, allowing them to access global markets without the complexities of a full cross-listing (Hitt et al., 2006). Some firms may target foreign investors by participating in global investor roadshows, listing on regional trading platforms like the OTC markets, or issuing American Depositary Receipts (ADRs), all of which can broaden their investor base and increase visibility in international markets (Karolyi, 2006).

Enhancing operational efficiency is another pivotal response. To compete effectively, firms may implement cost optimization strategies such as streamlining supply chains, renegotiating vendor contracts, or leveraging economies of scale (Porter, 1985). These measures not only improve profitability but also make the firm more appealing to investors. Investment in innovation is another avenue, where peers increase spending on research and development, adopt new technologies, or develop proprietary products and services to strengthen their competitive edge (Schumpeter, 1942). Additionally, operational improvements may extend to customer-facing processes, such as enhancing service quality, offering product customization, or improving response times, fostering brand loyalty and driving revenue growth (Prahalad & Hamel, 1990).

By adopting these strategies, peers aim not only to counterbalance the competitive advantage of the cross-listed firm but also to position themselves as credible and appealing alternatives to investors and customers. These efforts, if successful, can elevate industry standards as a whole, creating a ripple effect of enhanced governance, efficiency, and market integration.

In the short term, the announcement of a peer's cross-listing may result in positive or negative abnormal returns for non-cross-listed peers. Some studies suggest that peers might benefit from a "spillover effect," as the cross-listing enhances the visibility of the entire industry, leading to increased investor interest in the sector (Lang, Lins, & Miller, 2003). On the other hand, the cross-listing firm's signaling of superior governance and market positioning may highlight the deficiencies of its peers, resulting in negative abnormal returns for them (Ferris, Kim, & Noronha, 2002).

In the long term, the effects on peers' returns depend on their ability to adapt to the new competitive landscape. Peers that successfully emulate the cross-listed firm's strategies may experience improved financial performance and legitimacy, aligning their returns with those of the

cross-listed firm. Conversely, peers that fail to adapt may suffer from diminished market share, reduced investor confidence, and lower valuations (Chen, Su, & Zheng, 2016).

The collective response of peers provides insights into the broader trends of market globalization. If peers actively pursue strategies similar to the cross-listed firm—such as improving governance or seeking internationalization—it suggests a broader shift toward globalization in the local market. Conversely, if peers remain static, it indicates resistance to globalization and reliance on localized market practices.

Competitive dynamics theory helps to contextualize these strategic responses by highlighting the interdependence of firms within the same competitive domain. The cross-listing event triggers a dynamic interplay of actions and reactions, influencing both individual firm performance and the overall trajectory of the local market.

Hypothesis 4a: *In the short term, peers experience negative abnormal returns as they respond to the cross-listing firm's announcement, which highlight competitive disadvantages.*

Hypothesis 4b: *In the long term, peers that adopt competitive strategies in response to the cross-listed firm will see improved returns, while those that fail to adapt will experience reduced performance and valuation.*

D) Sample Selection and Data

This study examines the impact of cross-listing on the financial integration, stock returns, and market risk profiles of foreign firms listed on major U.S. exchanges, specifically the NYSE, AMEX, and NASDAQ, between 1990 and 2023. The data were gathered from multiple financial databases, including WRDS, CRSP, Compustat, Bloomberg, and Thomson Reuters Financial & Risk. The final sample comprises 1,142 companies from 44 countries that pursued cross-listing during the study period.

The cross-listing process for the NYSE, AMEX, and NASDAQ involves distinct timelines and procedures. For the NYSE and AMEX, foreign firms must file a formal application, typically announced publicly within a week after submission. For NASDAQ-listed firms, the announcement occurs electronically upon application approval. Unlike the NYSE and AMEX, which require a confidential preliminary review to assess eligibility, NASDAQ applications undergo no such review. Once the application is approved, firms coordinate with the respective exchange to determine their listing date.

For this study, firms that became cross-listed on U.S. exchanges for the first time were selected. To ensure robust and accurate analysis, firms with missing or incomplete data for stock returns and financial metrics over critical estimation and observation periods were excluded. Additionally, firms from industries that were inconsistent with prior studies were removed to maintain comparability and relevance.

Monthly stock return data for the cross-listed firms were collected for a 96-month window surrounding the cross-listing date: 60 months before the listing ($t = -60$ to $t = -1$) served as the estimation period, and 36 months after the listing ($t = 0$ to $t = +36$) formed the observation period. For the estimation period, data were used to calculate baseline metrics, such as beta and expected

returns, while the observation period facilitated the analysis of abnormal returns and changes in risk exposure. Event study methodology was employed to examine the financial impact of cross-listing, with the event date denoting the month the security started trading on a U.S. exchange.

The data sources for this research include multiple verified repositories. Listing dates for the NYSE, AMEX, and NASDAQ firms were obtained directly from the exchanges' official records and verified using secondary sources such as historical stock price records. Return data for each cross-listed firm were retrieved from CRSP and complemented with global market indices from Bloomberg. Peer firms were selected from the same industry and home country as the cross-listed firms, with market values comparable to the sample firms, ensuring a meaningful benchmark for peer analysis. Short-term T-bill rates from the Federal Reserve and corresponding international central banks served as the risk-free rate for return calculations.

In addition to stock returns, key financial and governance variables were collected from Compustat and Bloomberg. These variables include market capitalization, book-to-market ratios, trading volume, and industry classifications.

E) Methodology

Suppose that cross-listing on a major U.S. exchange causes a change in the expected return on the stock from R^* before to R after the listing. Consistent with institutional theory, the hypothesis is that the enhanced legitimacy and increased investor confidence and decreased information asymmetry resulting from adherence to stricter U.S. regulatory standards will lower the firm's cost of capital and thus reduce its expected return.

$$H_0: R^* - R = 0$$

$$H_1: R^* - R > 0$$

One method of testing this null hypothesis involves estimating R^* and R for individual securities immediately before and immediately after their cross-listing dates, respectively. However, careful attention must be paid to potential complications arising from institutional effects, peer signaling, and regulatory harmonization dynamics.

Cross-listing on a U.S. exchange not only signals a firm's commitment to higher governance standards but also aligns it with a more stringent institutional environment. According to institutional theory, such alignment enhances the firm's legitimacy, fostering trust among global investors. This institutional alignment is expected to create favorable conditions for the firm's stock performance, particularly around the cross-listing announcement date, as markets respond to the perceived benefits of operating within a stronger regulatory framework.

The timing of the cross-listing announcement is critical for hypothesis testing, as institutional effects and peer signaling are likely to be most pronounced during this period. Furthermore, the structural differences between U.S. and domestic markets, including disparities in regulatory rigor and investor protections, may amplify the positive institutional effects of cross-listing. This amplification can result in abnormal stock price reactions during the event window.

Selection bias may arise if the estimation period immediately prior to the cross-listing is used to estimate R^* . Prior research has noted that stock prices often exhibit upward trends leading up to

cross-listing announcements due to market anticipation of institutional alignment and signaling effects. Such bias can distort R^* , increasing the risk of Type I errors. To mitigate this, the methodology incorporates robust estimation windows that precede the pre-listing run-up period, ensuring a more accurate baseline.

To test for changes in expected returns, this study employs event study methodology with cumulative abnormal returns (CARs) and utilizes a Capital Asset Pricing Model (CAPM)-based return-generating process. Persistence of abnormal returns in the post-listing period, coupled with the theoretical framework of institutional legitimacy, will serve as evidence of the long-term financial integration benefits associated with cross-listing.

E1. Examining Abnormal Returns Around Cross-Listing Announcements

To test **Hypothesis 1a** and **Hypothesis 1b**, which explore the dynamics of abnormal returns around the cross-listing announcement, we employ the event study methodology. This approach is widely used in financial research to evaluate the impact of specific events on stock prices by isolating abnormal returns from expected market performance.

For each firm i in period t , the return-generating process is expressed as:

$$R_{it} = \alpha_i + \beta_i R_{mt} + \epsilon_{it} ,$$

Where, R_{it} is the return of the stock for firm i in period t , R_{mt} is the market return during the same period, α_i is the intercept representing the average abnormal return, β_i is the market risk factor (sensitivity to market returns) and ϵ_{it} is the residual error term.

The parameters α_i and β_i are estimated using ordinary least squares (OLS) over a 60-month pre-event estimation window ($t = -60$ to $t = -1$).

Abnormal returns (AR_{it}) are calculated as the difference between the actual stock return and the expected return predicted by the Capital Asset Pricing Model (CAPM):

$$AR_{it} = R_{it} - (\alpha_i + \beta_i R_{mt}) ,$$

To analyze the market reaction at both the firm level and across the sample, we compute Average Abnormal Returns (AAR_t) and Cumulative Abnormal Returns ($CAR(a, b)$):

$$AAR_t = \frac{1}{N} \sum_{i=1}^N AR_{it} ,$$

$$CAR(a, b) = \sum_{t=a}^b AAR_t ,$$

where N is the number of firms in the sample, a and b define the start and end of the observation window.

We examine abnormal returns over multiple windows to capture both short-term and medium-term effects around the announcement date ($t=0$). These windows include:

- Short-term: $[-3, +3]$, $[-5, +5]$, and $[-11, +11]$ days around the announcement,

- Medium-term: [-1, +1], [-3, +3] month,
- Long-term: [-1, +1], [-3, +3] years.

To evaluate the significance of the abnormal returns t-tests are used on AAR_t to determine whether the average abnormal returns significantly differ from zero. Cumulative abnormal returns $CAR(a, b)$ are tested for statistical significance using the formula:

$$t(CAR(a, b)) = \frac{CAR(a, b)}{\sqrt{(b - a + 1) \cdot \sigma^2}}$$

where σ^2 is the variance of the abnormal returns during the estimation period.

To ensure the robustness of our findings we account for potential liquidity and signaling effects that may influence short-term abnormal returns. selection bias is mitigated by using a sufficiently long estimation period to accurately calculate baseline parameters α_i and β_i .

By employing this methodology, we isolate the impact of cross-listing announcements on firm stock prices and test Hypotheses 1a and 1b:

- **Hypothesis 1a** posits that before the cross-listing, firms experience high information asymmetry and low legitimacy, leading to higher expected returns, which may result in positive abnormal returns.
- **Hypothesis 1b** posits that after the cross-listing, enhanced legitimacy and reduced information asymmetry lead to a closer alignment between expectations and actual returns, potentially resulting in lower or even negative abnormal returns.

E2. Methodology of Segmented Markets vs Non-Segmented Markets

Classification of Segmented and Non-Segmented Markets

To categorize countries into segmented and non-segmented markets, this study employs the Chinn-Ito Financial Openness Index (KAOPEN). The KAOPEN index, derived from International Monetary Fund (IMF) data, measures a country's degree of capital account openness based on the presence of multiple types of capital controls, such as restrictions on capital inflows and outflows (Chinn & Ito, 2006). Markets are classified as segmented if their KAOPEN index score indicates significant capital controls and regulatory constraints, suggesting a higher "liability of foreignness" (Zaheer, 1995). Conversely, non-segmented markets are characterized by lower institutional barriers, greater capital account openness, and more established reputations, making them more integrated into the global financial system (Hoskisson et al., 2000; Li, 2003).

Methodology for Comparing Abnormal Returns Across Market Types

To examine the differential impact of cross-listing on firms from segmented versus non-segmented markets, this study utilizes an event study methodology to calculate abnormal returns both before and after the cross-listing announcement.

Before Cross-Listing:

For firms from segmented markets, the expectation is that they face higher information asymmetry and lower legitimacy, resulting in a larger gap between expected and realized returns compared to firms from non-segmented markets.

For each firm i in a segmented market, the abnormal return (AR_{it}) is calculated using the Capital Asset Pricing Model (CAPM) as:

$$AR_{it} = R_{it} - (\alpha_i + \beta_i R_{mt}) ,$$

Where, firms from non-segmented markets are expected to show a smaller gap, given their more favorable operating conditions and lower barriers to international acceptance.

After Cross-Listing:

Following cross-listing, firms from segmented markets are expected to reduce this gap as they align with international standards, enhancing their legitimacy and reducing information asymmetry. This leads to lower abnormal returns as market expectations align more closely with actual performance.

For firms from non-segmented markets, no significant change in abnormal returns is expected after cross-listing, as these firms are already operating under more transparent and less segmented market conditions.

The abnormal returns are similarly calculated using the CAPM model as described above.

Comparative Analysis of Abnormal Returns: To test Hypotheses 2a and 2b, we compare the calculated abnormal returns of segmented and non-segmented market firms both before and after their respective cross-listing events. For Hypothesis 2a, comparing the abnormal returns before the cross-listing involves a t-test for independent samples to assess whether the mean abnormal returns differ significantly between segmented and non-segmented firms. The formula used is:

$$t = \frac{AR_{segmented} - AR_{non-segmented}}{\sqrt{\frac{s^2_{segmented}}{n_{segmented}} + \frac{s^2_{non-segmented}}{n_{non-segmented}}}}$$

Where $AR_{segmented}$ and $AR_{non-segmented}$ are the mean abnormal returns for firms from segmented and non-segmented markets, respectively, $s^2_{segmented}$ and $s^2_{non-segmented}$ are the variances of the abnormal returns in these groups and $n_{segmented}$ and $n_{non-segmented}$ are the number of firms in the segmented and non-segmented groups, respectively.

For Hypothesis 2b, which deals with post-cross-listing abnormal returns, a similar t-test is employed to compare the mean abnormal returns of segmented and non-segmented firms after the cross-listing event.

E3. Methodology for risk exposure

To examine Hypothesis 3a and Hypothesis 3b regarding changes in a firm's market beta before and after cross-listing, the following methodology is employed:

Estimating the Market Beta

The first step involves estimating the beta of each firm relative to its home country market index before and after the cross-listing event. The pre-cross-listing period spans a 60-month estimation window ($t = -60$ to $t = -1$), capturing the firm's risk exposure tied primarily to its local market. During this period, a firm's beta is calculated as:

$$R_{it} = \alpha_i + \beta_{home,i} R_{home,t} + \epsilon_{it} ,$$

where R_{it} is the return of the stock for firm i in period t , $R_{home,t}$ is the home market return during the same period, α_i is the intercept, $\beta_{home,i}$ represents the firm's sensitivity to the home market, and ϵ_{it} is the residual error term.

Following cross-listing, the post-event period extends to 36 months ($t = 0$ to $t = +36$). The beta is re-estimated during this period to reflect any changes due to the cross-listing:

$$R_{it} = \alpha_i + \beta_{home,i} R_{home,t} + \epsilon_{it}$$

In addition to the home market beta, each firm's beta relative to the U.S. market is estimated for both pre- and post-cross-listing periods to capture its global risk exposure.

Comparing Pre- and Post-Listing Betas

To test for significant changes in a firm's risk profile due to cross-listing, the difference in beta values is computed for each firm relative to both its home market and the U.S. market. For the home market beta:

$$\Delta\beta_{home,i} = \beta_{home,i,post} - \beta_{home,i,pre}$$

A negative change in $\Delta\beta_{home,i}$ indicates a decrease in the firm's dependency on local market conditions post cross-listing. Similarly, for the U.S. market:

$$\Delta\beta_{U.S,i} = \beta_{U.S,i,post} - \beta_{U.S,i,pre}$$

A positive change in $\Delta\beta_{U.S,i}$ reflects an increase in the firm's exposure to global risk factors after cross-listing.

Statistical Analysis

To test for significant differences in these beta changes between segmented and non-segmented firms, t-tests or regression models are employed. Specifically, the mean changes across all firms within each group are compared:

$$t = \frac{\Delta\beta_{home,segmented} - \Delta\beta_{home,non-segmented}}{\sqrt{\frac{S^2_{home,segmented}}{n_{segmented}} + \frac{S^2_{home,non-segmented}}{n_{non-segmented}}}}$$

$$t = \frac{\Delta\beta_{U.S,segmented} - \Delta\beta_{U.S,non-segmented}}{\sqrt{\frac{s^2_{U.S,segmented}}{n_{segmented}} + \frac{s^2_{U.S,non-segmented}}{n_{non-segmented}}}}$$

where $\Delta\beta$ represents the mean change in beta, s^2 is the variance of the changes, and $n_{segmented}$ and $n_{non-segmented}$ are the number of firms in the segmented and non-segmented groups respectively.

Robustness Checks

To ensure the robustness of the findings, cross-validation is performed by splitting the sample into different subsamples based on industry or firm characteristics. Re-estimating betas and testing changes across these subgroups allows for validation of the results. Sensitivity analyses are also conducted using different estimation windows around the cross-listing event (e.g., ± 6 months) to confirm the consistency of the observed changes in beta.

E4. Methodology for Analyzing Peer Effects

Selection of Peers

To study the peer effects arising from a firm's cross-listing, we first identify peers within the same industry that share a closer market capitalization with the cross-listed firm. This ensures that comparisons are made between firms of similar size and within comparable competitive environments. Peers are selected from the same industries as the cross-listed firms, ensuring homogeneity in industry characteristics, which is crucial for a meaningful comparison. The selection criterion is based on both industry classification and market capitalization proximity, as firms with closer market caps are more likely to be directly influenced by the strategic actions of the cross-listed firm.

Calculation of Abnormal Returns

For each cross-listed firm, we calculate its abnormal returns around the announcement date of its cross-listing. Abnormal returns are derived by subtracting the expected returns (based on a market model) from the actual returns on the event days. This method captures the direct market reaction to the news of cross-listing, isolating the effect from overall market movements.

Subsequently, we also calculate the average abnormal returns for the identified peers around the same announcement date. This process yields 1142 abnormal returns for cross-listed firms (we have 1142 cross-listed firm in our sample) and 1142 corresponding average abnormal returns for their peers across different event windows. This is done over various short-term windows: [-3, +3], [-5, +5], and [-11, +11] days and long-term event windows: [-1, +1] year and [-3, +3] years, relative to the announcement.

To statistically compare these returns across peers and cross-listed firms. The formula used for this t-test is:

$$t = \frac{AR_{cross-listed} - AR_{peer}}{\sqrt{\frac{s^2_{cross-listed}}{n_{cross-listed}} + \frac{s^2_{peer}}{n_{peer}}}}$$

Here, $AR_{cross-listed}$ and AR_{peer} represent the mean abnormal returns for the cross-listed firms and their peers, respectively. $s^2_{cross-listed}$ and s^2_{peer} are the variances of these returns, and $n_{cross-listed}$ and n_{peer} are the number of firms in each group. This statistical test helps identify significant differences in abnormal returns between peers and cross-listed firms, providing insights into how the cross-listing influences competitive dynamics in the short term.

In the long term, the focus shifts to evaluating the sustainability of peer responses to the competitive pressures created by the cross-listing event. Comparisons are made over longer event windows: [-1, +1] year and [-3, +3] years post-announcement. To explore this relationship, a regression analysis is employed where the dependent variable is the abnormal returns of peers post-cross-listing. The regression model is specified as follows:

$$AR(peer)_i = \alpha + \beta_1 \times PostCrossListing + \beta_2 \times PeerResponse + \epsilon_i$$

In this model, $AR(peer)_i$ denotes the abnormal returns of peers, $PostCrossListing$ is a dummy variable representing the period after the cross-listing, and $PeerResponse$ captures the extent of peer adaptation to the cross-listed firm's strategies. α is the constant term, while β_1 and β_2 are coefficients to be estimated. ϵ_i is the error term.

A regression model is employed to determine the relationship between the changes in abnormal returns of peers and the adaptation strategies they may adopt in response to the cross-listing firm. Variables considered include improvements in corporate governance, internationalization efforts, operational efficiency enhancements, and adoption of new technologies. The model helps capture the extent to which peers are able to mirror the cross-listed firm's strategies and its impact on their financial performance and market positioning. This regression analysis enables the assessment of whether peers that successfully adapt to the cross-listed firm's strategic moves experience improved returns over time, as hypothesized in hypothesis 4b.

F) Conclusion and Discussion

This study revisits the impact of cross-listing on major U.S. exchanges, focusing on its implications for financial integration, stock returns, and risk profiles of international firms. The findings indicate that cross-listing significantly lowers the cost of capital for firms, leading to a reduction in expected returns and negative abnormal returns in the short and medium term following the announcement. This aligns with institutional theory, which posits that adherence to stricter U.S. regulatory standards enhances legitimacy and reduces information asymmetry, thereby attracting more confident investor behavior and lowering risk premiums.

Furthermore, firms from non-segmented markets are found to benefit more from cross-listing due to their existing lower barriers to international operations, which facilitates easier integration into

global financial standards. These firms experience greater alignment with international norms, leading to less volatility in stock returns and improved long-term financial performance.

The study also highlights the peer effects associated with cross-listing, revealing that the listing of a firm influences the financial outcomes of its industry peers. Peers respond strategically to a cross-listing firm's enhanced visibility and market position, adopting similar governance structures, internationalization strategies, and operational efficiencies.

G) Future Research Considerations

Future studies could focus deeper into the effects of cross-listing by examining the specific characteristics of firms that might moderate its impact, such as ownership structure, firm size, or the specific home country regulatory environment. Additionally, research could explore the long-term sustainability of these benefits and examine whether they extend beyond financial performance to include operational and strategic dimensions. A comparative analysis across different geographical regions or emerging markets could further enhance our understanding of the universal versus context-specific benefits of cross-listing.

Another promising avenue for future research is to investigate the role of cultural factors in the cross-listing process. Cultural alignment between home and host countries might significantly influence the success of cross-listing, impacting investor perception and strategic adaptations. Moreover, the impact of technological advancements and the digitalization of financial markets could be integrated into the analysis, offering insights into how firms leverage new technologies to maintain competitive advantages post-cross-listing.

In conclusion, while this study sheds light on the immediate financial benefits and peer interactions associated with cross-listing, it also opens up multiple avenues for future research to build on these findings.

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